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Description - The theory of the RF/Pen cabinet operation.

1- ANALOG PROCESSOR BOARD

1-1 Overview

The analog processor board (APB) performs the redundant power monitor envelope detection and scaling functions, the linearization of the MR power amplifier with envelope feedback (EFB II), sums the base band spectroscopy power monitor signal into the head A/D port, and interfaces the MR and spectroscopy amplifier control bus with the Communications Manager Board power monitoring functions.

1-2 Band Pass Filters

Output signal Body_BPF_A is input signal Body_In_A band filtered between 10 and 90 MHz.

Output signal Head_BPF_A is input signal Head_In_A band pass filtered between 10 and 90 MHz.

SVC_Out is the combined Body_BPF_A and Head_BPF_A signals after sampling with the directional couplers in block Band Pass Filters.

EFB_RF_FB is the combined Body_Out and Head_Out signals after sampling with the directional couplers. This section also incorporates the EFB gain adjustment potentiometers accessible on the connector panel for field calibration of RF power. The potentiometers adjust the attenuation introduced into the EFB feed back path. The EFB controlled gain of the MR amplifier is proportional to the attention.

1-3 Envelope Detection

Output signal Body_Env_Det_A(B) (Head_Env_Det_A(B)) is the detected, actively filtered, and scaled envelope of input signal Body_BPF_A(B) (Head_BPF_A(B)). The second order infinite gain low pass multiple feedback filter has a pass band of 18.5 kHz. The last operational amplifier provides a potentiometer for factory use to scale the Body_Env_Det_A(B) (Head_Env_Det_A(B)) signal to full scale compatibility with the A/D converters located on the Communications Manager Board.

1-4 Interlocks

The interlock circuit will force the respective Body_Env_Det_A(B) (Head_AD_A(B)) output port to full scale when the dc resistance looking out of the corresponding input port Body_In_A(B) (Head_In_A(B)) is lower than 25 ohms, or greater than 104 ohms.

The LC circuit on port Body_In_A(B) (Head_In_A(B)) isolates RF from the interlock and prevents loading of the RF inputs.

A voltage divider is used to set reference voltages for the inputs of two comparators. The second comparator input is formed by a second divider where the resistance to ground is formed by the internal resistance in the directional coupler on the rear of the narrow band amp. The second dividers voltage is nominally .682 Vdc for a dc resistance of 50 ohms, 1.29 Vdc for 104 ohms, and .357 Vdc for 25 ohms.

If the second dividers voltage falls below or above the comparators reference voltage, one of the comparators outputs will go to a logic low, saturating the following PNP, turning on the LED, and pulling the Body_Env_Det_A(B) (Head_Env_Det_A(B)) line to a voltage one diode drop below the power supply voltage.

1-5 Spectroscopy Detectors

The spectroscopy receivers amplify the Spectro_In_A(B) signals with an adjustable linear gain ranging from 0 to 51 and sums the result with the Head_Env_ Det_A(B) signals to form the Head_A/D_ A(B) output signals.

1-6 Envelope Feedback

Two modes of operation are available with envelope feedback, preset and enabled. In preset mode, the EFB enable signal disconnects the integrator circuit and fixes the control voltage of the VCA (voltage controlled attenuator) circuitry to a constant dc voltage.

Conversely, in enabled mode, the error signal from the log amps is connected to the integrator and the MR amplifier gain is controlled by the attenuation in the feedback path. (Caution: A minimum level of RF signal on EFB_RF_REF is necessary for the control loop to operate properly).

The logarithmic amplifiers remove the RF carrier and compress the envelope of both the reference and feedback signals. To minimize the error in the correction EFB provides for the MR amplifier, it is imperative that the log amps are matched over the specified dynamic range. Two factory adjustment potentiometers are provided to match the Vo/Vi slope and intercept response for a given RF.

The signals from the logarithmic amplifiers are differentially summed and the resulting error signal is integrated to control the VCA circuitry.

The VCA circuitry consists of an anti-log amplifier, four quadrant multiplier, and a differential voltage feedback operational amplifier gain block. See Illustration L3181A.

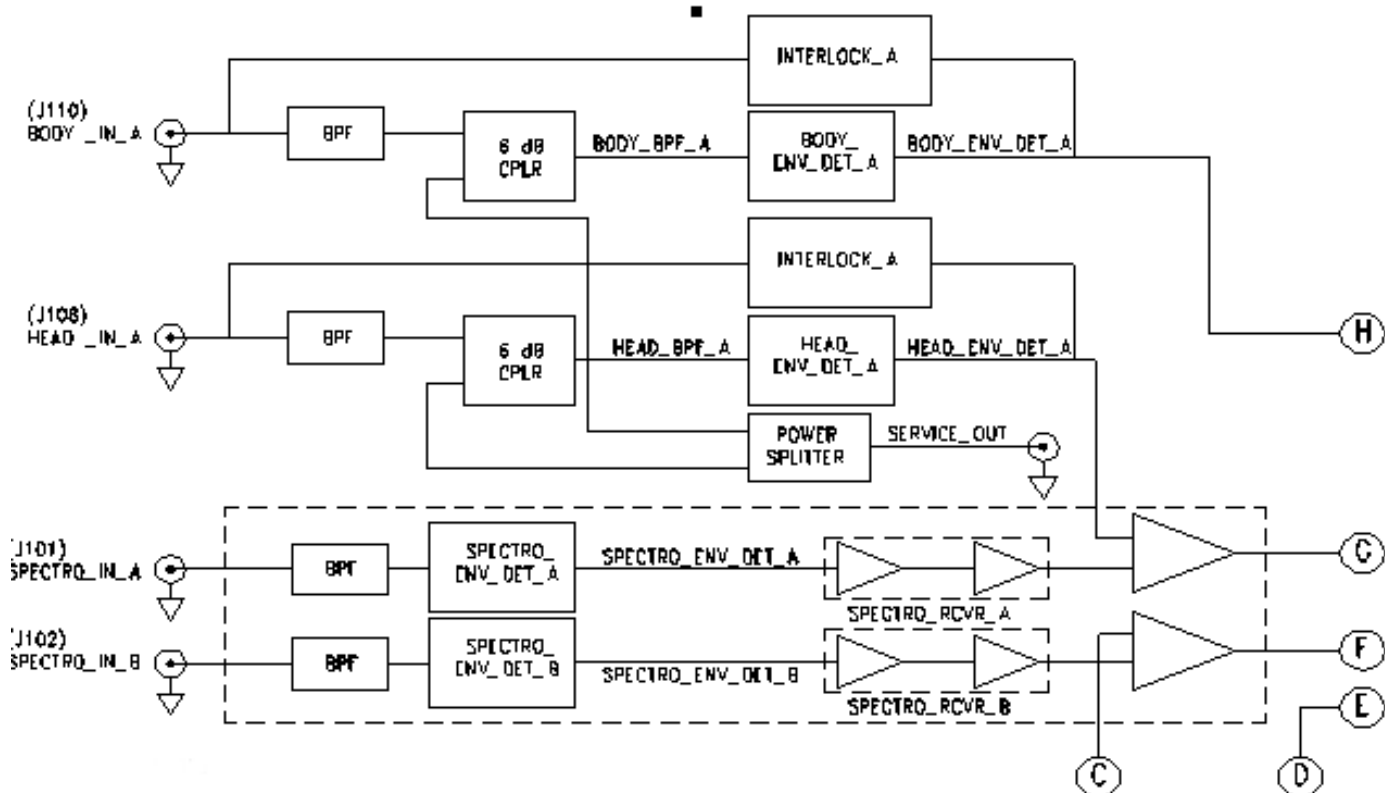


ILLUSTRATION L3181A
APB FUNCTIONAL BLOCK DIAGRAM - SECTION 1

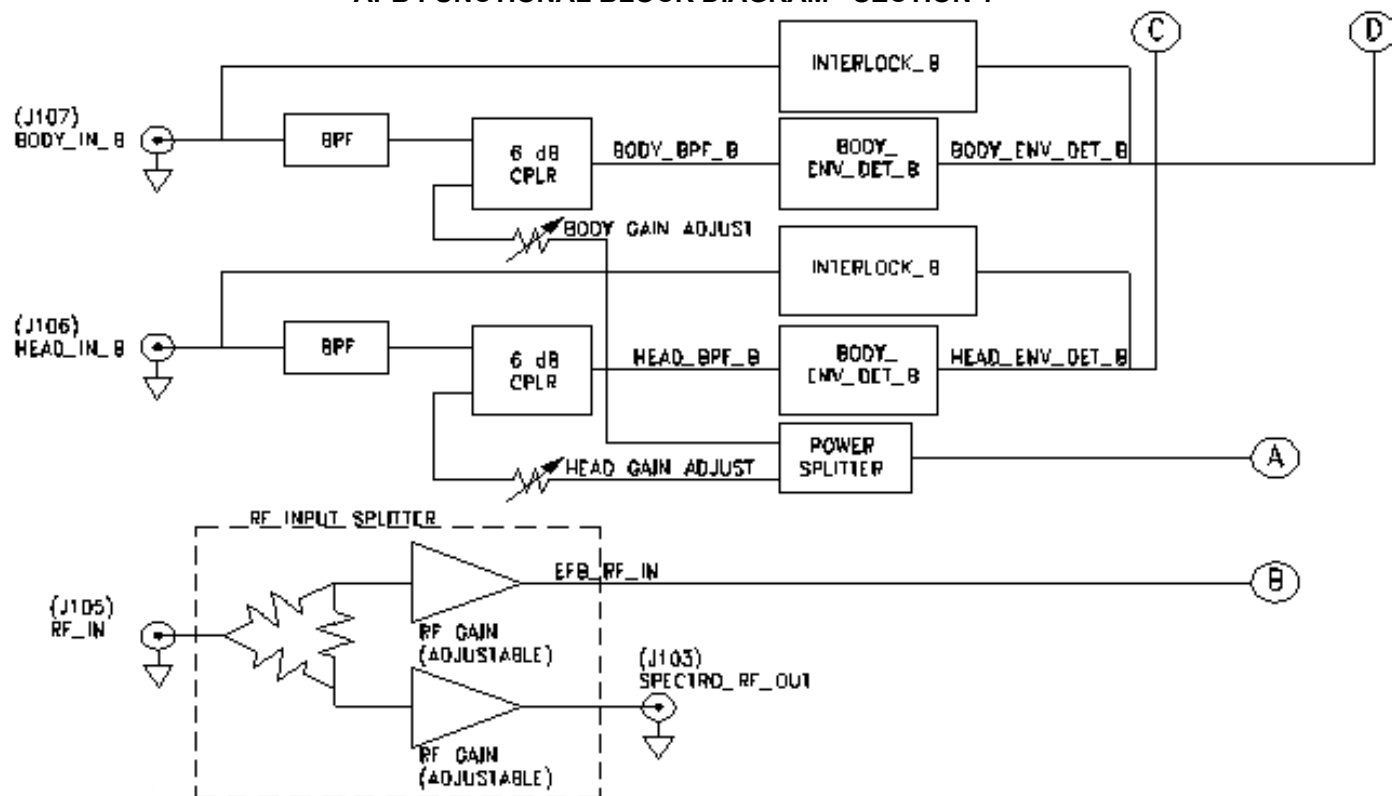


ILLUSTRATION L3181A
APB FUNCTIONAL BLOCK DIAGRAM - SECTION 2

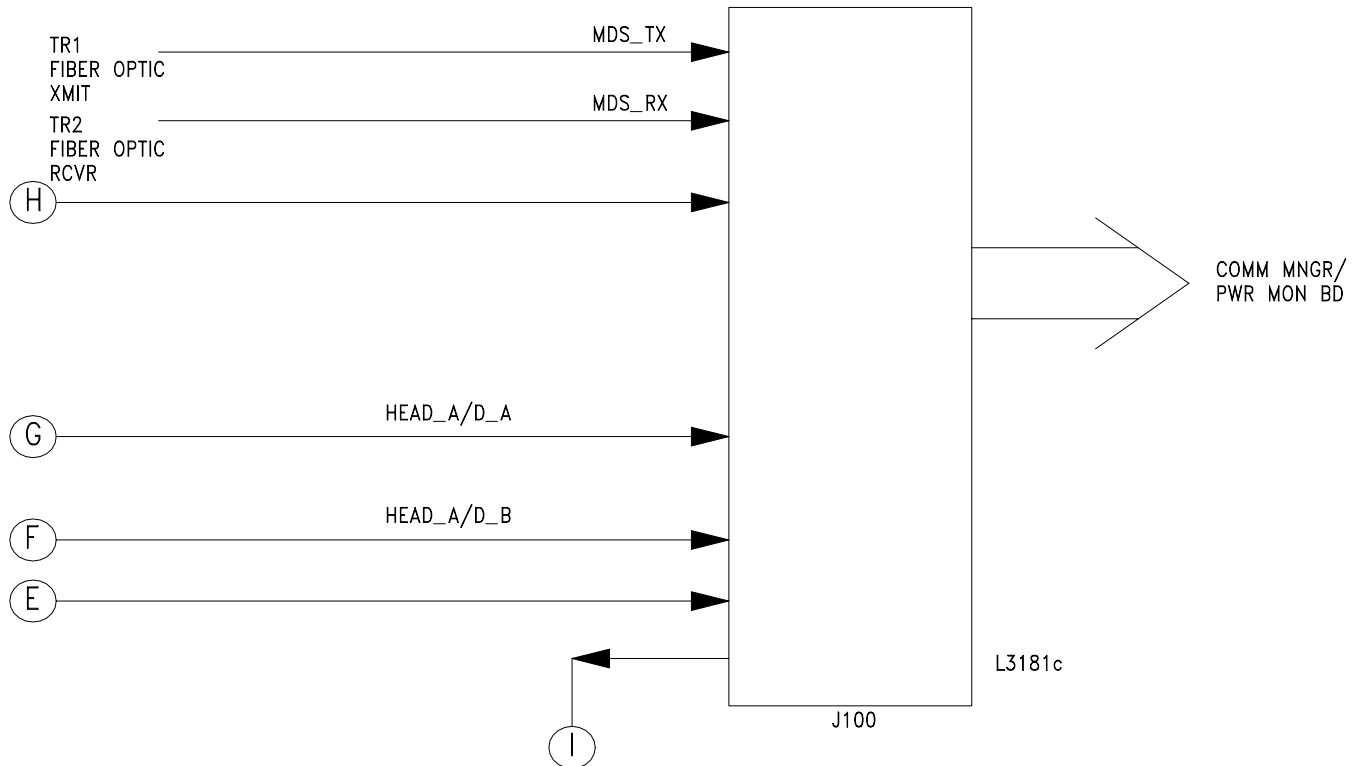


ILLUSTRATION L3181A
APB FUNCTIONAL BLOCK DIAGRAM - SECTION 3
APB FUNCTIONAL BLOCK

DIAGR

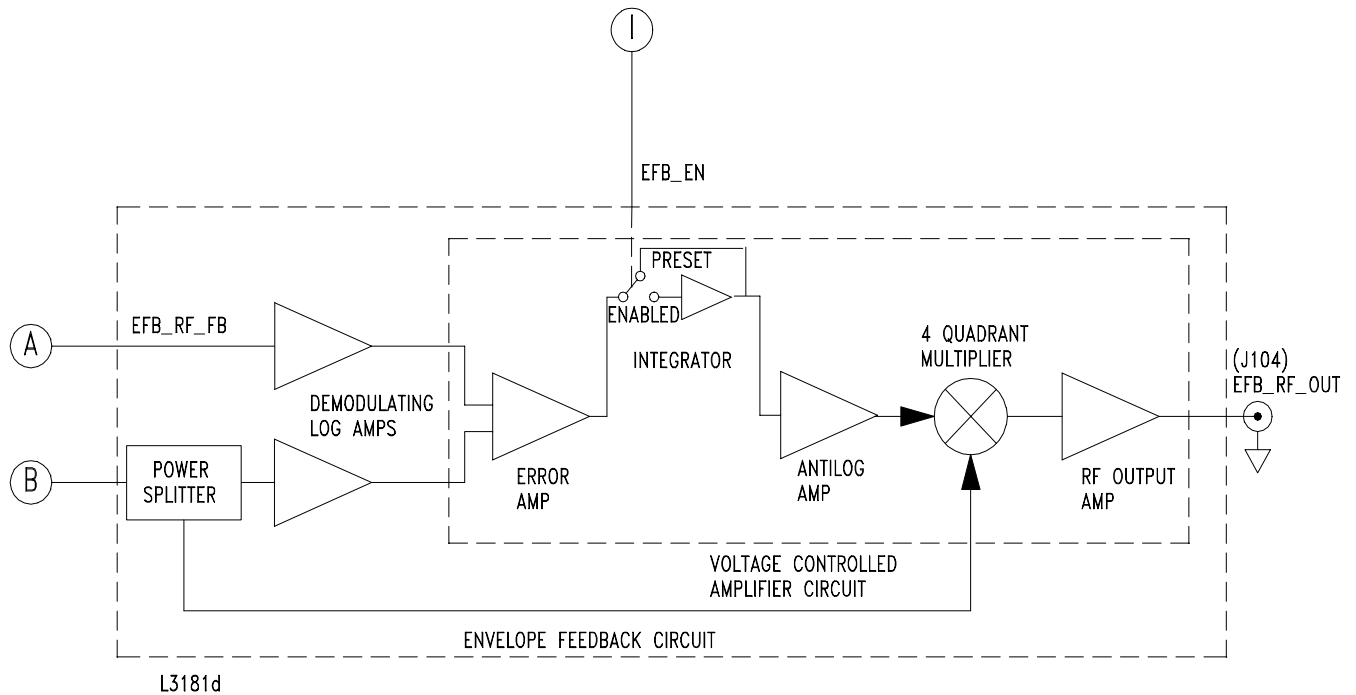


ILLUSTRATION L3181A
APB FUNCTIONAL BLOCK DIAGRAM - SECTION 4

NOTES:

2- COMMUNICATIONS MANAGER / POWER MONITOR BOARD

2-1 Overview

The Communications Manager/Power Monitor Board is responsible for 2 distinct digital tasks: It provides the interface between components of the RF/Pen cabinet and the controlling computers. It also contains two independent RF power monitors which help to Ensure that the amplifiers are being used properly. These two functions were contained on separate PCBs in earlier versions of the RF/Pen cabinet.

The controlling computer can present its commands over either of two ports: the MDS optical port or an auxiliary RS232 port (henceforth called the Aux port). These ports can be accessed independently. There is no need to synchronize the commands that are transferred over the two ports. Commands that are received over a given port are echoed back to the same port, and any responses that are due will be given to that port. Command collisions are sorted out within the Communications Manager. The command syntax varies between the two ports; however, all standard commands can be executed from either port.

All commands are targeted to some function within the RF cabinet. The functions that are accessible are:

Communications with the RF amplifiers.
Communications with the PIN drivers.
Communications with the RF monitors.

Note

The MDS port is a 9-bit RS-232 port transferred over optical cables. The actual optical connectors and the circuitry which converts to electrical signals are located on the APB board. The discussion of the MDS port which follows describes both the optical portions (on the APB board) as well as the electrical signals (on the Communications Manager/Power Monitor Board).

2-2 MDS optical

The incoming optical cable connector is located on the APB board. The optical signal is converted inside the connector into an electrical signal that can be viewed at TP28 (on the APB board). The signal should comply with the MDS specifications described in other GE documentation. In general, the signal encodes data at 9600 baud, 9 data bits, no parity, 1 stopbit, in a manner very similar to conventional RS-232. When the optical light is lit, the electrical signal should be digital low. This signal is the input to the SCI receiver pin on the HC11 microprocessor, U36 (on the Communications Manager/Power Monitor board).

The receive signal also drives a timer circuit which, in turn, resets the microprocessor. This provides the capability for the optical port to reset the microprocessor by sending a "light on" condition. The length of time required to cause reset is about 10 msec.

The outgoing optical cable connector is also located on the APB board. Outgoing data originates at the microprocessor (electrical signals) and is conducted (via PCB traces) across the APB board to the optical cable connector. The conversion to optical takes place within the connector.

The processor circuit also contains a switch that allows (or inhibits) incoming MDS data to be directly copied to the outgoing MDS transmit line. The normal state of the switch allows this path. Consequently, when MDS data are received, they are simultaneously echoed through the transmit pin. The switch is opened when the Communications Manager is a) responding to a MDS command, or b) originating a fault message.

2-3 Auxiliary serial port

The auxiliary serial port is a standard RS-232 port configured in the DCE pin-out. Consequently, a "null modem" cable is required to connect to standard computers. The port is fixed (not user configurable) at 9600 baud, no parity, 8 data bits, 1 stop bit.

The command set which is currently supported follows:

BRxx	Bicycle read address xx
BWxx,yy	Bicycle write address, data yy
SRxxxx	SPC read address xxxx (hc11 address space)
SWxxxx,yy	SPC write address xxxx, data yy
P1string	Copy string to pmon1 and return the response
P2string	Copy string to pmon2 and return the response
Mraddr	Amplifier read address addr
MWaddr data	Amplifier write address, data

All other commands are syntax errors, and cause the message "Syntax Error" to be returned to the interface. Most of the commands cause the construction and transmission of a serial string to one (or more) of the other serial ports. The communications over these ports is rigorously checked. Any error which is noted causes the message "Comm error" to be returned to the Aux port.

2-4 Pmon1 and Pmon2

Each of the two power monitors are identical in functionality. They use the same microprocessor circuitry and operate the same firmware. They each monitor the same signals, and report their findings in the same way. However, their hardware is completely independent of each other. They use separate power supplies and separate ICs.

Each power monitor has three dedicated jumpers that control its operation.

Pmon1

JP1: Connected = Normal,
 Not Connected = BODY
 Envelope disconnected
 JP2: A = Normal, B = Service
 JP3: B = Normal, A = Continuous self test
 JP4: A = Normal, B = Watchdog
 disabled
 TP3: BODY Envelope detect
 TP4: HEAD_A/D

Pmon2

JP5: Same as JP4
 JP6: Same as JP2
 JP7: Same as JP3
 JP9: Same as JP1
 TP10: HEAD_A/D
 TP12: BODY Envelope detect

The key on the front of the RF system controller chassis has three positions: Mode 2 (bypass), Mode 1 (Normal), and Reset. The signals from this key switch connect through the PIN Switch Driver Board and the associated cables to the power monitors.

JP8: A = normal, B = Amplifier fault 85 disabled

2-5 Diagnostics

Step 1: Ensure correct connection of the MDS port.

MDS: unplug the optical cable from the APB board and look into the optical cable. The command protocol requires that the controlling computer provides a periodic request for cabinet status. This should be visible as a brief pulse of red light in the cable, which occurs approximately once every 1.6 seconds. If this periodic pulse is not visible, check the MDS source and/or the optical cables.

When the optical cable is disconnected, the status request command cannot reach the Communications Manager. The cabinet is expecting this periodic status request. When the request fails to come within approximately two seconds, an MDS BROKE error condition is assumed and a message is sent accordingly. The response to this error condition can be seen as a brief pulse of red light in the optical transmit connector. This pulse should occur at the error time-out rate (approximately once every two seconds).

If the pulsing red lights are visible in both the connector and in the cable, the MDS hardware is working correctly. Connect the optical cables to the optical connectors. The transmitter (the cable with the pulsing light) should be connected to the receiver (the connector without the pulsing light).

Step 2: Ensure correct operation of the MDS port.

Using an oscilloscope, examine test points TP27 and TP28 on the APB board. These are the MDS receive and transmit data from the Communications Manager. Ensure that the commands are correct.

Step 3: Ensure correct operation of the Aux port.

Connect a serial cable between a computer (the standard laptop will do) and the cabinet connector J46. Configure the communications to 9600 baud, No parity, 8 data bits, 1 stop bit. Send any command and look for the response. A good command to send is the Enter key (ASCII 0x0d) alone. The Enter key terminates all Aux port commands. This command alone is a syntax error and should return a suitable message.

Step 4: Ensure correct operation of the NB amplifier serial port.

Cause a command to be sent to the amplifier which causes a visible response. There are two reasonable ways to do this:

Turn off the cabinet circuit breaker. All power should be off. Disconnect the optical receive cable (its okay to disconnect both optical cables). Turn on the cabinet circuit breaker. The comm manager should quickly recognize the MDS BROKE condition. One of the responses to this condition is to send the "O1" command to the amplifier. Watch the amplifier; it should begin its power up sequence. If so, the port is working correctly.

Using a serial computer (like the laptop), send a legal serial command to the amplifier. At the present time, only the MR and MW commands have been decoded. Send the command "MR 3011" to read the amplifier status byte. Send the command "MR 3012" to read the amplifier mode echo and so forth. All characters should be echoed by the amplifier and all syntax should be checked rigorously. If there is any doubt, connect the serial cable directly to the amplifier itself and type the same commands. In both cabling situations, MR commands should respond with the appropriate data.

Step 5: Ensure the correct operation of the RF monitor serial ports.

The power monitors are located on the Communications Manager / Power Monitor Board itself. Consequently, this test is entirely internal. Using either of the two control ports (MDS or Aux), send a legal command string to each of the power monitors. Be careful, there are two possible monitor types which can be found on

Communications Manager boards: the 1.5T and the 1.0T. The command string which is sent to the monitors contains a byte which indicates the monitor type. When a legal string is sent to a monitor, the monitor responds by returning a signature of its type. The signature for a 1.0T monitor is "F0" and for a 1.5T configured with envelope feedback is "3C". Send the string and look at the response. If the command string is legal, the monitor will return the correct response. Otherwise, it will return nothing. Ensure that the monitors can be accessed correctly.

Step 6: Ensure correct operation of the PIN Switch Driver interface.

Using either of the two control ports (MDS or Aux) send a command to the PIN Switch Driver Board and Ensure that the command was received. There are many ways to accomplish this task. The easiest way is to recognize that the multicoil option circuitry is included on all cabinet PIN Switch Driver Boards. The circuitry is enabled by an 8-bit latch (4 bits enable the drivers, 4 bits enable the multicoil fault detection PALs) decoded at SCM address 0x29. This circuitry is initialized at reset by hardware to 00h (disable all drivers and all faults). The SCM read address (0x29) has also been decoded so that software can determine the current contents of the latch.

In order to ensure proper operation communications with the PIN Switch Driver Board, write the values 0x01, 0x02, 0x04, 0x08, ... 0x80 to SCM address 0x29. For each write, read the data back and compare.

Step 7: Ensure correct operation of the Spectro amp interface. See Illustration L3183A.

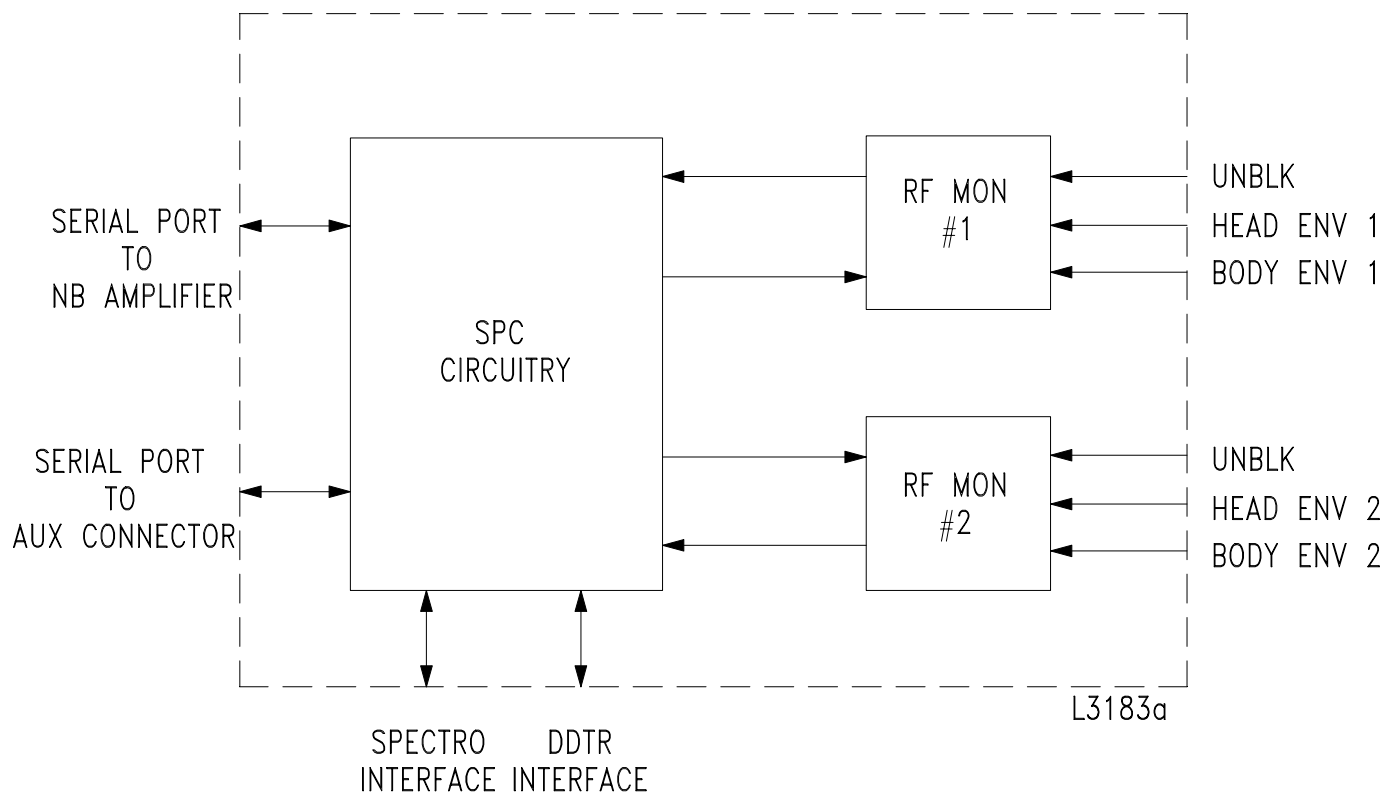


ILLUSTRATION L3183A
COMMUNICATION MANAGER/POWER MONITOR BOARD FUNCTIONAL BLOCK DIAGRAM

3- PIN SWITCH DRIVER BOARD

3-1 Overview

The purpose of the PIN Switch Driver Board is to individually select the different system coils, and to provide the distinction between transmit and receive modes on these coils by biasing PIN diodes located at the coils to either pass or prohibit the flow of RF energy.

Pulsed drive signals (amplifier unblank signals) are directed to the various drive elements; multicoils, T/Rs, and body drivers, as specified by the Communications Manager/Power Monitor Board in the RF System Controller (RFSC). Additionally, open circuit and short circuit error detection signals for each of the drivers are sent to Programmable Logic Arrays which are read by the Communications Manager/Power Monitor Board.

3-2 Multicoil Drivers

The four multicoil drives are calibrated for output level accuracy ($\pm 5V \pm 0.1V$) via R129 and R133 sensed at TP2, 14, 20, and 26. Additionally, these pots calibrate window comparator references VREFA and VREFB for error detection purposes.

Drive control of the multicoil drivers is established by PAL U70 if JP57-60 pins 5 and 6 are connected, or is manually selected for test purposes if the JP57-60 jumpers are connecting pins 1 and 2 (+5V multicoil drive) or pins 3 and 4 (-5V multicoil drive).

High power operational amplifiers U6, U18, U34, and U48 are configured for a gain of -1 relative to the drive levels from U22 and U28. These amplifiers are specified to supply $\pm 5V$ at a rated current of 1 amp. Sense resistors R6, R41, R89, and R118 control both the output current limiting of the amplifiers (fed back across pins 5 and 8) as well as the open/short detection signal levels.

The drivers nominally run at a current of 0.50 Amps into rated loads. A short circuit condition is considered to be currents exceeding 0.55 Amps, while an open circuit condition is considered to be currents less than 0.45 Amps. Precision op amps (U9, U20, U39, and U53), amplify the voltage resulting from the operating current through the sense resistors by a factor of approximately 7.2. This voltage, appearing at test points TP7, 11, 22, and 28, is then fed to window comparators U8 and U42 to sense open and short circuit conditions. When the amplified sense voltage rises above VREFA (while the amplifier is sourcing (+5V)), an open circuit condition is signaled. When the voltage falls below VREFB (while the amplifier is sinking (-5V)), a short circuit condition is signaled. The open circuit detection signals feed PAL U70 while the short circuit detection signals feed PAL U64.

3-3 T/R Drivers

The three T/R drivers are for Body, Head, and Spectro modes. They are designed to supply $+4 \pm 0.1V @ 4$ Amps (Body), $+8V \pm 0.1V @ 2.5$ Amps (Head), or $+6V \pm 0.1V @ 2.5$ Amps (Spectro). Calibration of the desired drive voltage level is accomplished via potentiometers R26 (Head), R13 (Body), and R18 (Spectro), while monitoring the output voltages at test point TP6, TP15, and TP5 respectively. During receive mode, the output will be clamped at approximately -12V and will sink 300 mA ± 50 mA into a rated load.

Drive control of the TR drivers is communicated via the Communications Manager/Power Monitor Board if jumper JP87 has pins 5 and 6 connected, or is selected for test purposes to be either transmit only (pins 3 and 4 connected) or receive only (pins 1 and 2 connected). As with the Multicoil drivers, the high power op amps (U5, U17, and U33) are configured to have a gain of -1 with respect to the drive signal.

Current limit of the output is accomplished in two ways. First, sense resistors R20, R62, and R106, wired across pins 5 and 8 of the high power op amps, define the maximum forward drive current during transmit mode and limit that value to approximately 5 Amps. Secondly, current limit resistors R8, R45, and R96 are enabled during receive mode only and will limit current draw to $-300 \text{ mA} \pm 50 \text{ mA}$.

Open circuit errors are specified to be generated when load currents falls below 0.5 Amps. Open circuit detection uses the corresponding voltage generated across the sense resistors R15, R57, and R101, amplified by a factor of 100 by precision amplifiers U1, U14, and U30. During transmit mode (sourcing positive voltage), when the drive current falls below 0.5 Amps, the resulting voltage (appearing at test points TP1, TP13, and TP18) is compared to an established reference level (+5V), is found to be lower than that reference, and the open circuit error signal is activated.

Short circuit detection works by sampling the sink voltage. When this voltage becomes more positive when compared to an established threshold of $-10\text{V} \pm 0.5\text{V}$, the short circuit signal is activated.

The open circuit and short circuit detection signals feeding PAL U32 are valid only during positive T/R drive (transmit) and negative T/R drive (receive) respectively.

3-4 Body 1, Body 2, and Direct Drive Drivers

These three drivers are required to either sink a specified current (2 Amps $\pm 200 \text{ mA}$ for Body 1 and Body 2, 1.1 Amps $\pm 100 \text{ mA}$ for Direct Drive) or to source either 500V $\pm 25\text{V}$ (1.0T) or 1000V $\pm 50\text{V}$ (1.5T).

The voltage source is U94 which will output 500V if JP1 pins 1 and 2 are connected or 1000V if JP1 pins 2 and 3 are connected. A safety interlock will break the connection at JP2, disabling the voltage source, in the event the RFSC lid is opened.

Drive control is communicated via the Communications Manager/Power Monitor Board if jumper JP5 has pins 1 and 2 connected, or is selected for test purposes to be either current sink mode or high voltage source mode via JP6.

When instructed by the drive control, the 500V/1000V voltage source is switched to the driver outputs via IGBT Q8. Transistor driver U102 biases Q8 either fully on or fully off utilizing a floating dc supply (T1, U101) of approximately ± 10 volts. When Q8 is not being driven on, IGBTs Q1, Q3, and Q6 are activated by the enable of switch U75, establishing a defined current sink loop. This current loop acts on the voltage appearing across sense resistors R162, R281, R200, R226 and R282, adjusting the current through the IGBTs to maintain a voltage drop of approximately 1 volt across the sense resistors. When switch U75 is disabled, transistors Q2, Q4, and Q5 are activated, shutting the current loop IGBTs fully off. RC time constants C164/ R249 and R236/ C150/ D28 ensure that the voltage sourcing IGBT and the current sinking IGBTs are not both on at the same time.

The short circuit error detection is to be sensed when sourcing high voltage. For short circuit detection ($20 \text{ mA} \pm 10 \text{ mA}$), the current flowing through optocouplers U61, U73, and U87 is sufficient to pull the optocoupler outputs down to a logic 0 when approximately 20mA flows through the driver output.

Open circuit error detection is sensed when sinking current and uses a field settable threshold selected by potentiometers R147, R207, and R228 (while monitoring test points TP19, 44, and 48) set to be 1 volt less than the voltage that appears at the driver outputs when in current sink mode at nominal load conditions. When the output voltage drops below this level, an open circuit condition is signaled. PAL U32 stores the error detect signals for the Direct Drive driver and PAL U56 stores the error detect signals for the Body 1 and Body 2 drivers.

3-5 Error detection timing

As specified in the detailed description of each of the drivers error detection methodology, open and short circuit conditions are detected or are valid only when the drivers are in certain drive states. The signal that defines each drivers state is schematically labeled as /UNBLNK. /UNBLNK passes through delay logic U78 and U79 which produces the delayed signal DUBLNK (approximately 250 ± 25 microseconds delayed from /UNBLNK). DUBLNK is responsible for clocking in all the driver error detection signals for interrogation by the Communications Manager/Power Monitor Board. All error detection signals are required to be stable within 175 microseconds from the appropriate edge of /UNBLNK to ensure DUBLNK clocks in accurate information.

U66 produces error signals indicating absence of the high voltage source or the on- board $\pm 15 \text{ V}$ regulated supply. These error signals are polled at a pre-defined rate by the system.

Switches SW1 and SW2 connecting to J5 contain error bypass and error reset functions for use by the Communications Manager/Power Monitor Board (see Illustration L3182A).

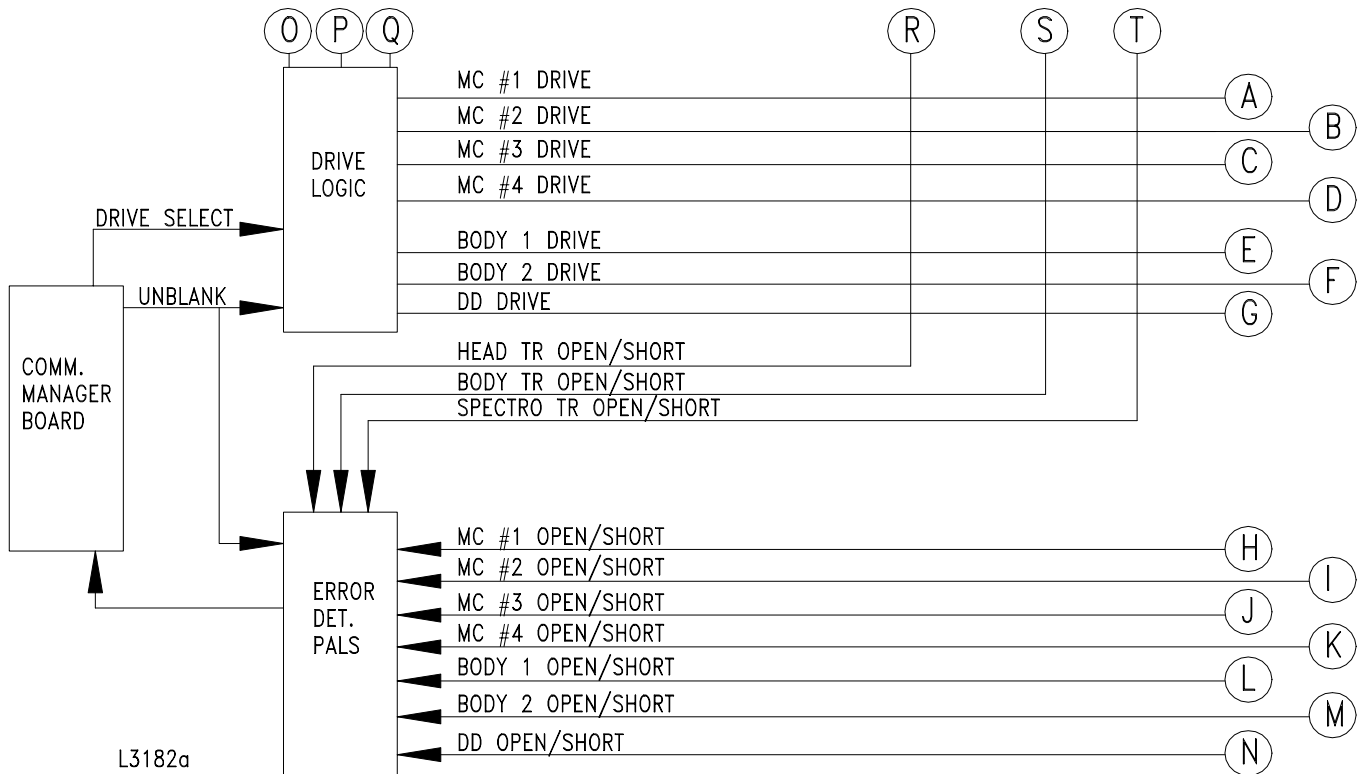


ILLUSTRATION L3182A
PIN SWITCH DRIVER BOARD FUNCTIONAL BLOCK DIAGRAM - SECTION 1

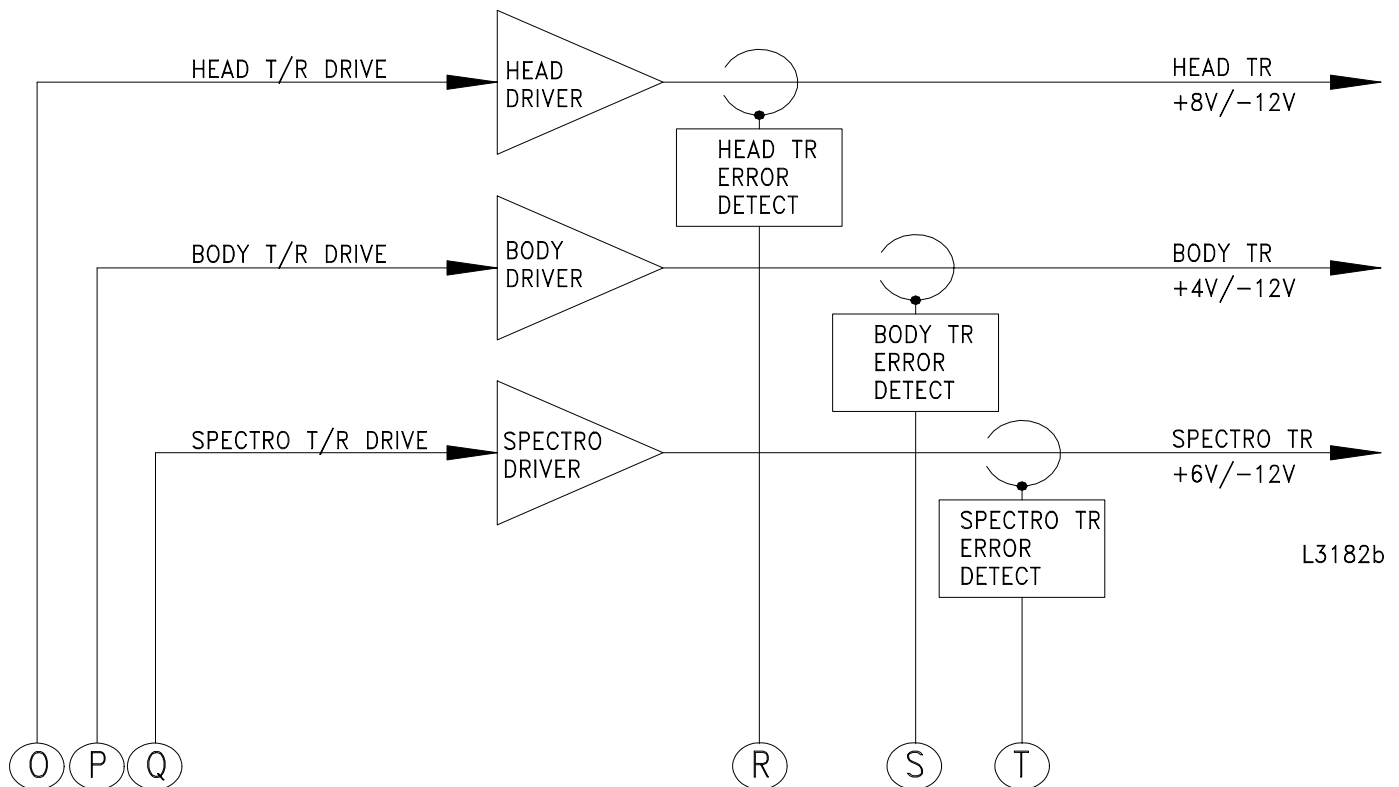


ILLUSTRATION L3182A
PIN SWITCH DRIVER BOARD FUNCTIONAL BLOCK DIAGRAM - SECTION 2

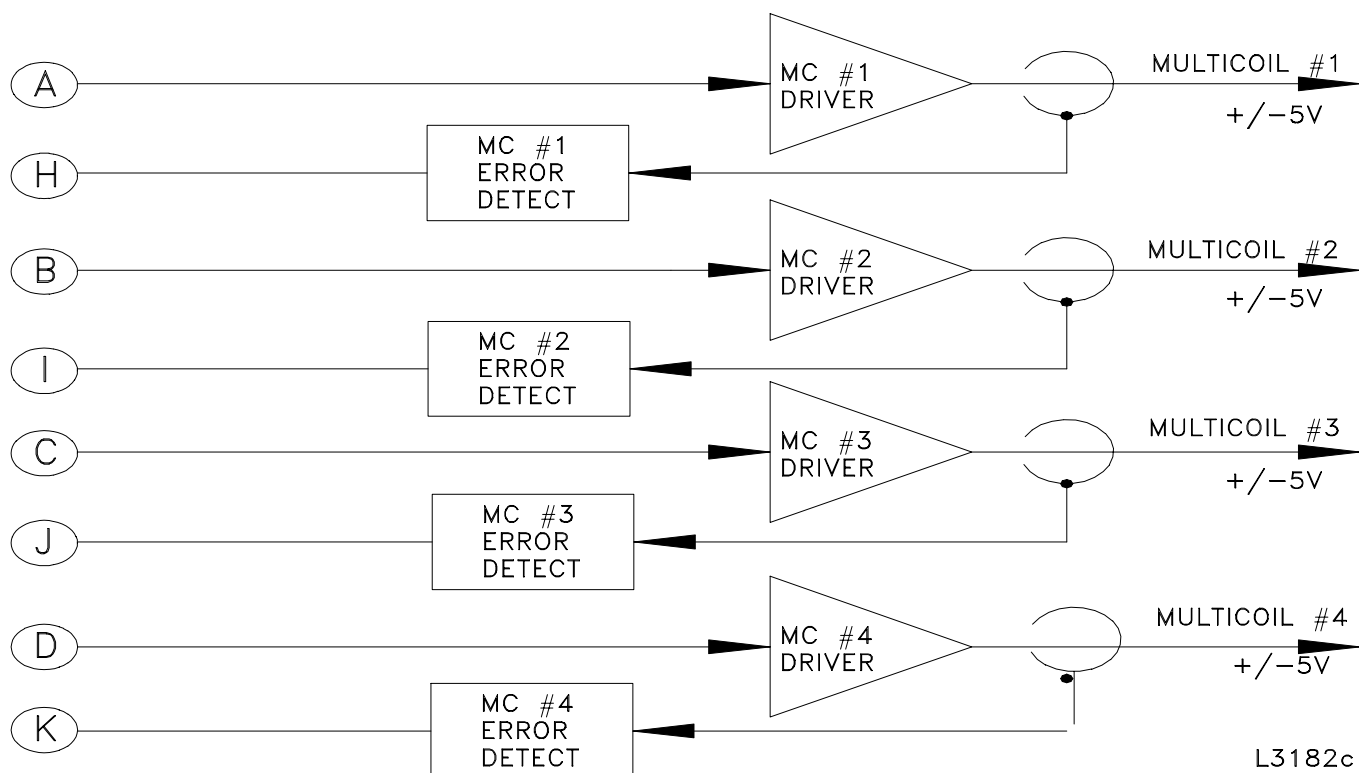


ILLUSTRATION L3182A
PIN SWITCH DRIVER BOARD FUNCTIONAL BLOCK DIAGRAM - SECTION 3

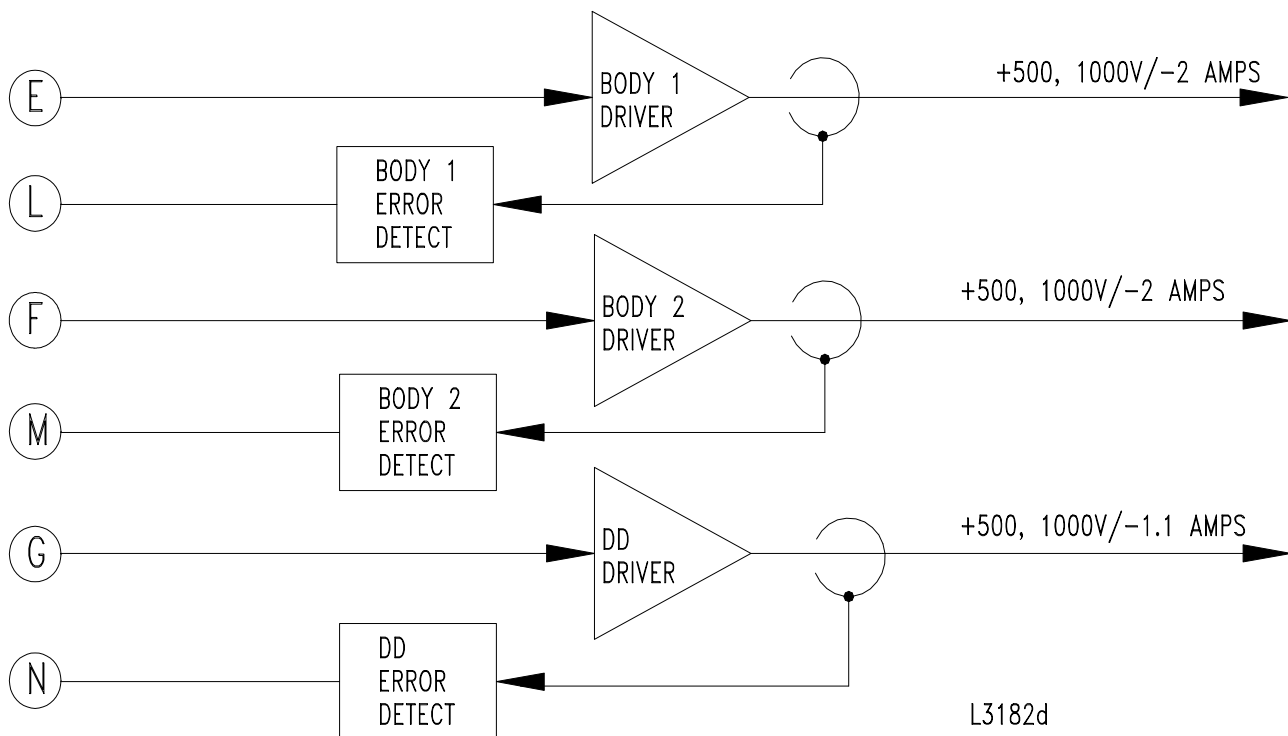


ILLUSTRATION L3182A
PIN SWITCH DRIVER BOARD FUNCTIONAL BLOCK DIAGRAM - SECTION 4

4- RF SYSTEM CONTROLLER POWER SUPPLY BOARD

The RF system controller power supply (RFSCPS) supplies the dc power for the functions of the RF system controller as well as fused ac for the fans in the RFSC, and for an isolated drive circuit on the PIN Switch Driver board. There are also LED indicators to show the status from the PIN Switch Driver board.

The +8 volt and ± 18 volt supplies are unregulated dc supplies which feed +5 volt and ± 15 volt regulators on the other RFSC PCBs. There are four distinct circuits, two +8 volt and two ± 18 volt. 120 Vac is fed into step-down transformers which in turn feed into bridge rectifiers. The output of the rectifiers is filtered by capacitors.

The ± 15 volt supply is regulated and is used for multicoil purposes outside of the RF/Pen cabinet. This supply is also generated from a rectified, stepped-down ac voltage.

The +24 volt supply is regulated and is used to power the 1000 volt power supply on the PIN Switch Driver Board. This supply is also generated from a rectified, stepped-down ac voltage (see Illustration L3184A).

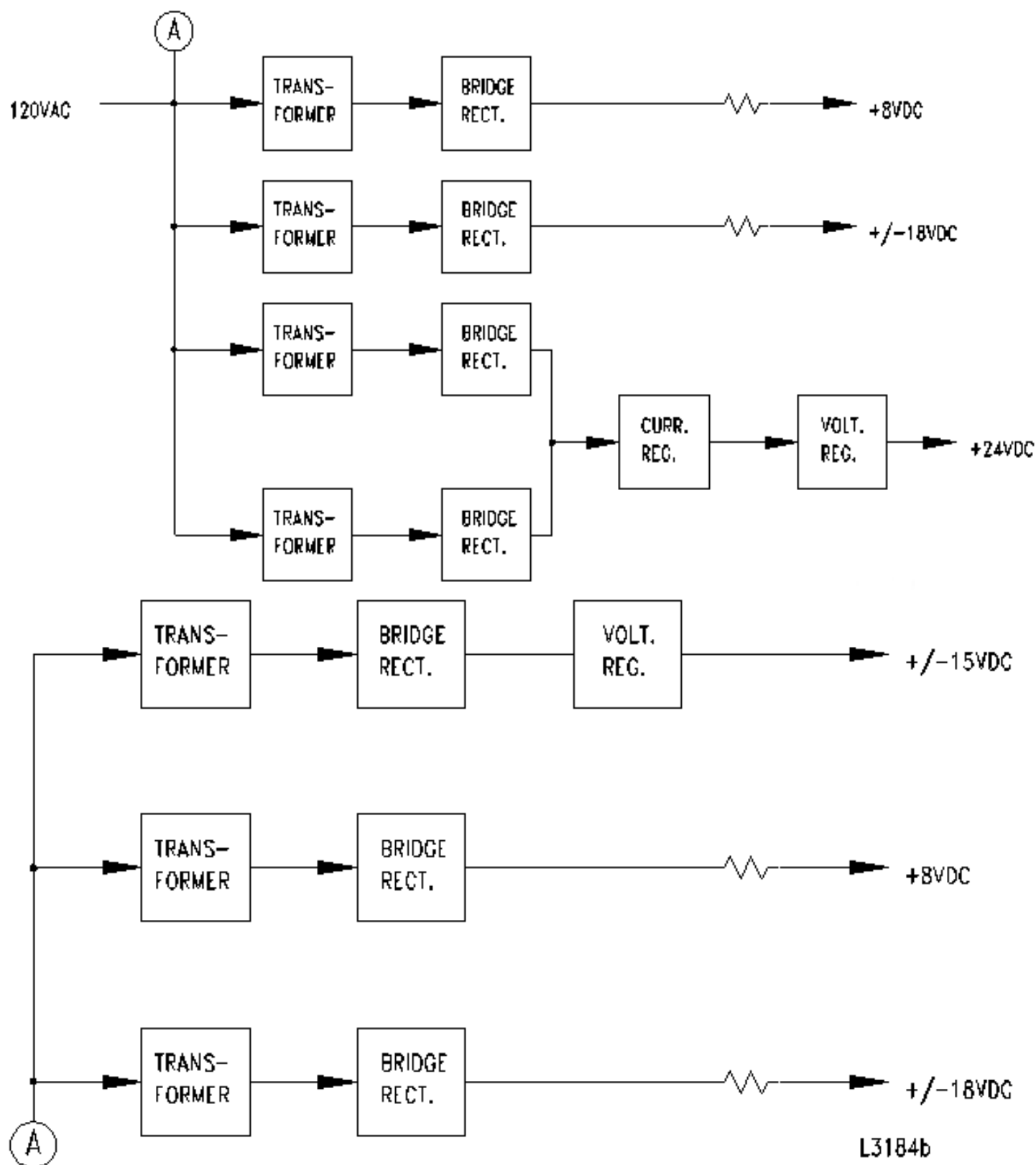


ILLUSTRATION L3184A
RFSC POWER SUPPLY BOARD FUNCTIONAL BLOCK DIAGRAM

5- MAGNET ENCLOSURE POWER SUPPLY (MEPS)

5-1 High Current Power Supply

The High Current Power Supply (HCPS) is comprised of the following power supplies:

1. PAL and Bore Light Power Supplies
2. Longitudinal/Docking Motor Power Supply
3. Pin Drivers Unregulated Power

The patient alignment light (PAL), and the bore light power supplies are nearly identical, so only one description is needed.

The unregulated voltage for the PAL supply is provided by one of the secondaries of the high current transformer (HCT). Three-phase secondary voltage from the HCT is coupled to a bridge rectifier with each phase protected by a fuse. Capacitors on the rectifier output provide smoothing and energy storage.

This unregulated voltage feeds three LM338 regulators in parallel. The adjustment pins of the regulators are driven by an LF412 op amp which controls the output voltage. When the RS422 input goes true, an exponential rise of +12.75V appears on the noninverting input of the LF412. With the output of the regulators fed back to the inverting input of the LF412, the op amp drives the adjustment pins of the regulators until the output reaches +12.75V and equalizes the inputs to the op amp.

When the RS422 input goes false, the voltage on the non inverting input drops to zero volts. The action of the op amp then forces the voltage at the inverting input to zero volts, and thus the output of the regulators.

The Longitudinal/Docking Motor power supply is a simple buck converter switching power supply.

The action of switching the IRFP250 FET at a high rate of speed causes the output voltage to be equivalent to the input voltage multiplied by the duty cycle of the FET switching signal.

The +15V is then fed into a switched capacitor voltage converter to provide an on board supply of -15V.

The +38.5V regulator uses a TL494 self-contained voltage mode switching regulator controller.

The remote sense plus voltage is returned to the voltage feedback divider, and the remote sense minus voltage is returned to TL494. Both of these voltages alter the voltage feedback to regulate +38.5V at the end of the remote sense connections. Over current protection is provided by the I-sense transistor and resistor. In the event that the load current goes to high, the voltage drop across the I-sense resistor causes the I-sense transistor to conduct and feed back a voltage to the TL494. This voltage operates the TL494 in a constant current mode. At the same time, this voltage is being sensed by the time delay comparator. If the condition exceeds a time period of approximately 200 msec, the comparator shuts down the regulator by pulling the dead time control pin to +5V.

The unregulated voltage for the Pin Switch Driver board is provided by two of the secondaries of the high current transformer. Three-phase secondary voltage from the HCT is coupled to full wave bridge rectifiers with each phase protected by a fuse. Capacitors on the outputs of the rectifiers provide smoothing and energy storage. The two unregulated voltages are routed up to the RF System Controller module for use by the Pin Switch Driver board.

The +24V needed to power the Longitudinal Drive Amp Relay is provided by an LM317HVT regulator.

The regulator is powered by the +38.5V Longitudinal/Docking supply. RS422 inputs control an H11A5100 optoisolator, which in turn controls a VN2222 FET in the return path from the relay.

5-2 Low Current Power Supply

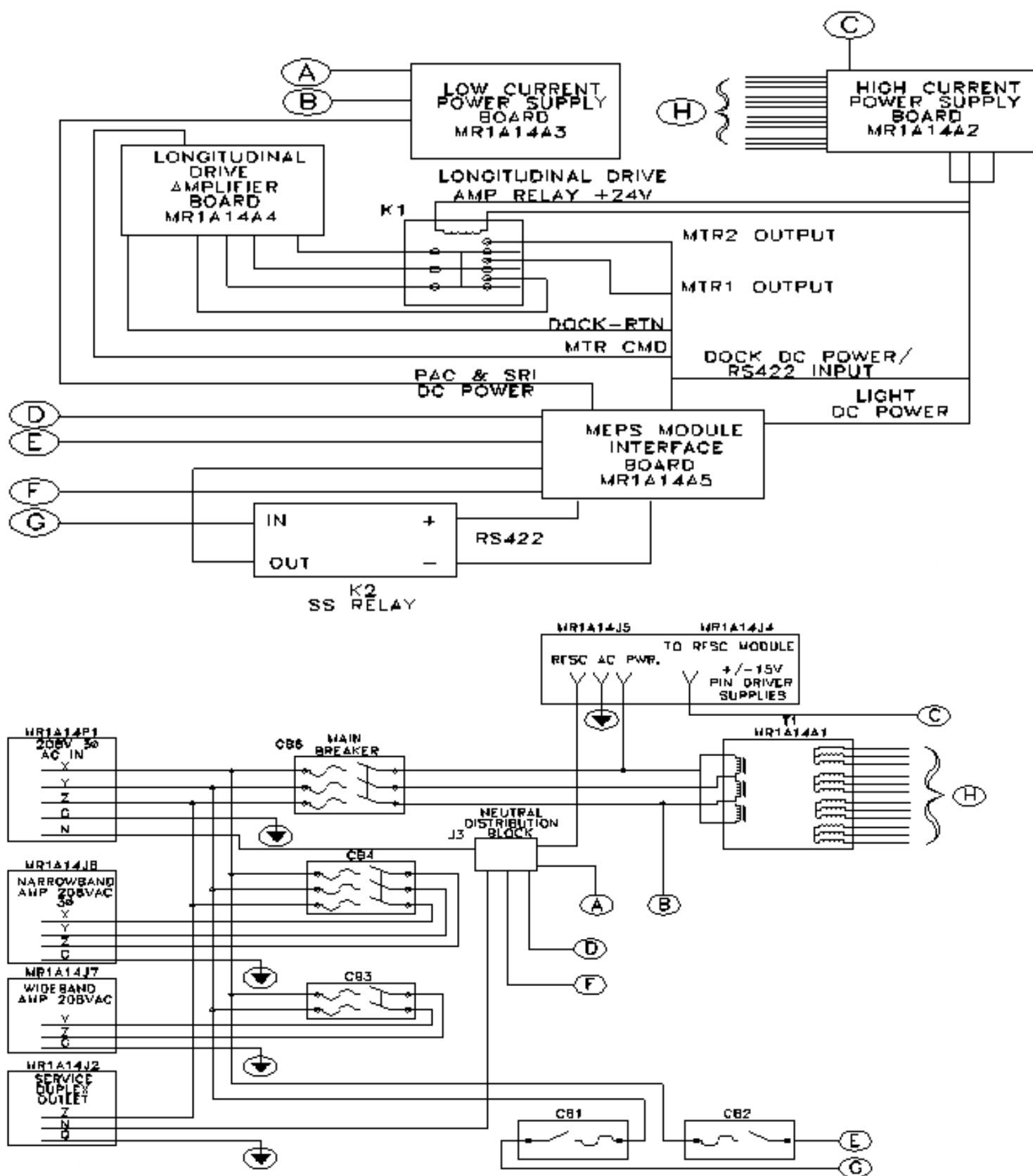
The low current power supply (LCPS) assembly is a collection of linear power supplies that meet the operating power requirements of the Physiological Acquisition Controller (PAC), and the Scan Room Interface (SRI). A breakdown of the supplies is indicated in Table 1.

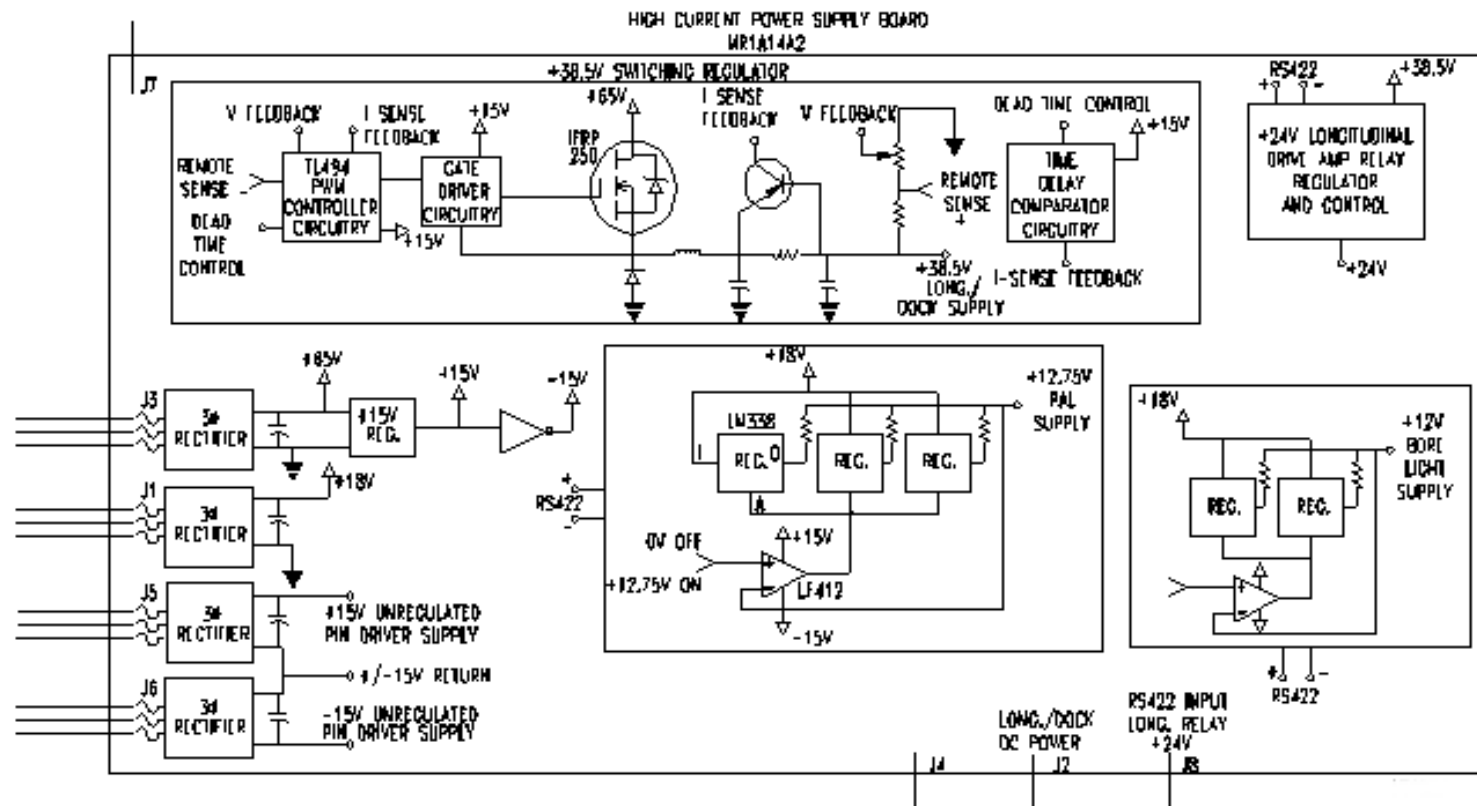
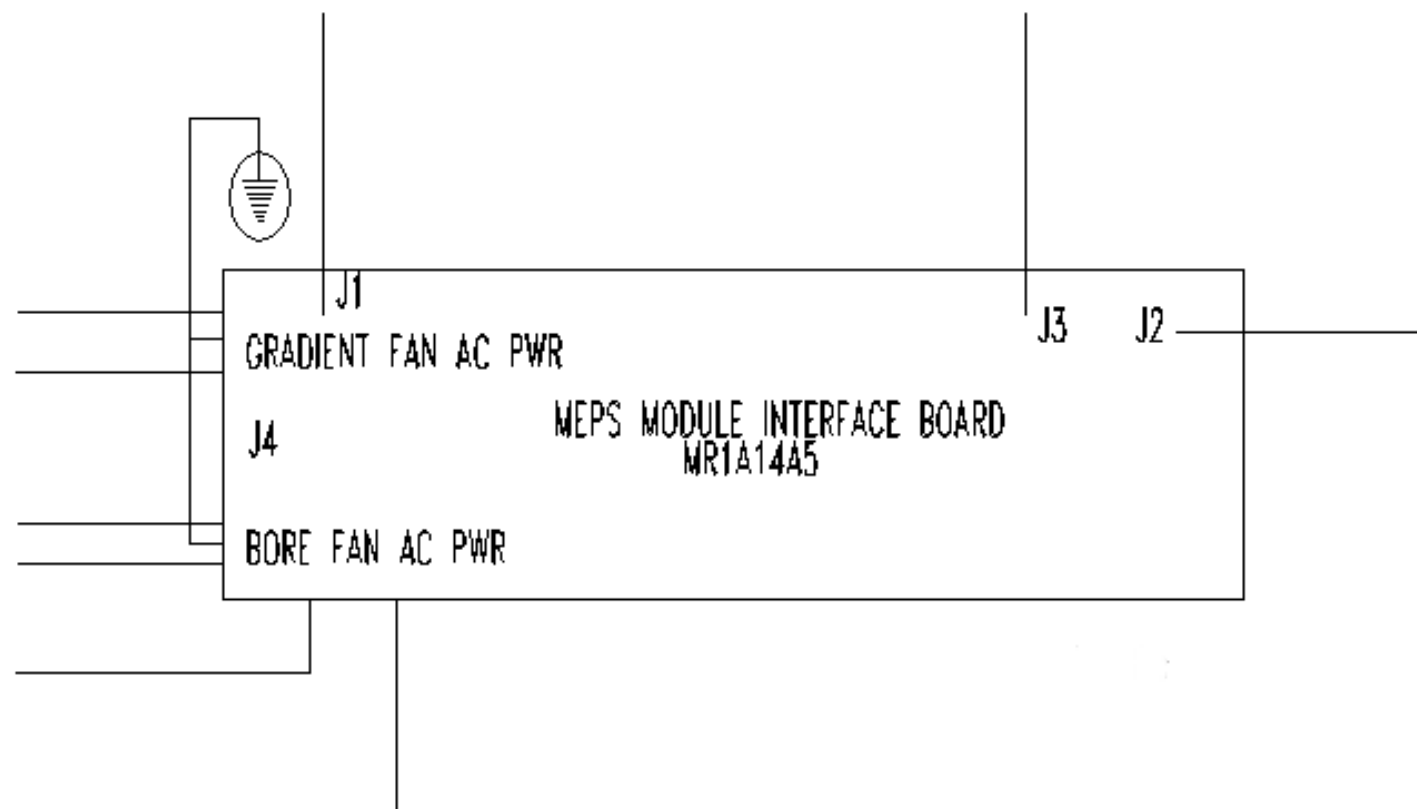
TABLE 1
LOW CURRENT POWER SUPPLY

PHYSIOLOGICAL ACQUISITION CONTROLLER	SCAN ROOM INTERFACE
15 Volts, 0.8 Amps	+24 Volts, 2.0 Amps
+9 Volts, 2.5 Amps	+12 volts, 2.0 Amps
-8 Volts, 0.4 Amps	+8 Volts, 2.0 Amps
-15 Volts, 0.4 Amps	

Note the SRI +8 volt supply with components F6, T6, and D6, etc. F6 (which is accessible from the front panel of the MEPS) protects transformer T6 from excessive temperature in the event of an excessive load on the transformer. Transformer T6 steps down the 120 Vac primary to a lower voltage, that when full-wave rectified by D6, and filtered by C22 provides a dc voltage that is sufficient to operate VR8, and yet not too high a value to cause VR8 to overdissipate.

See Illustration L3185A.





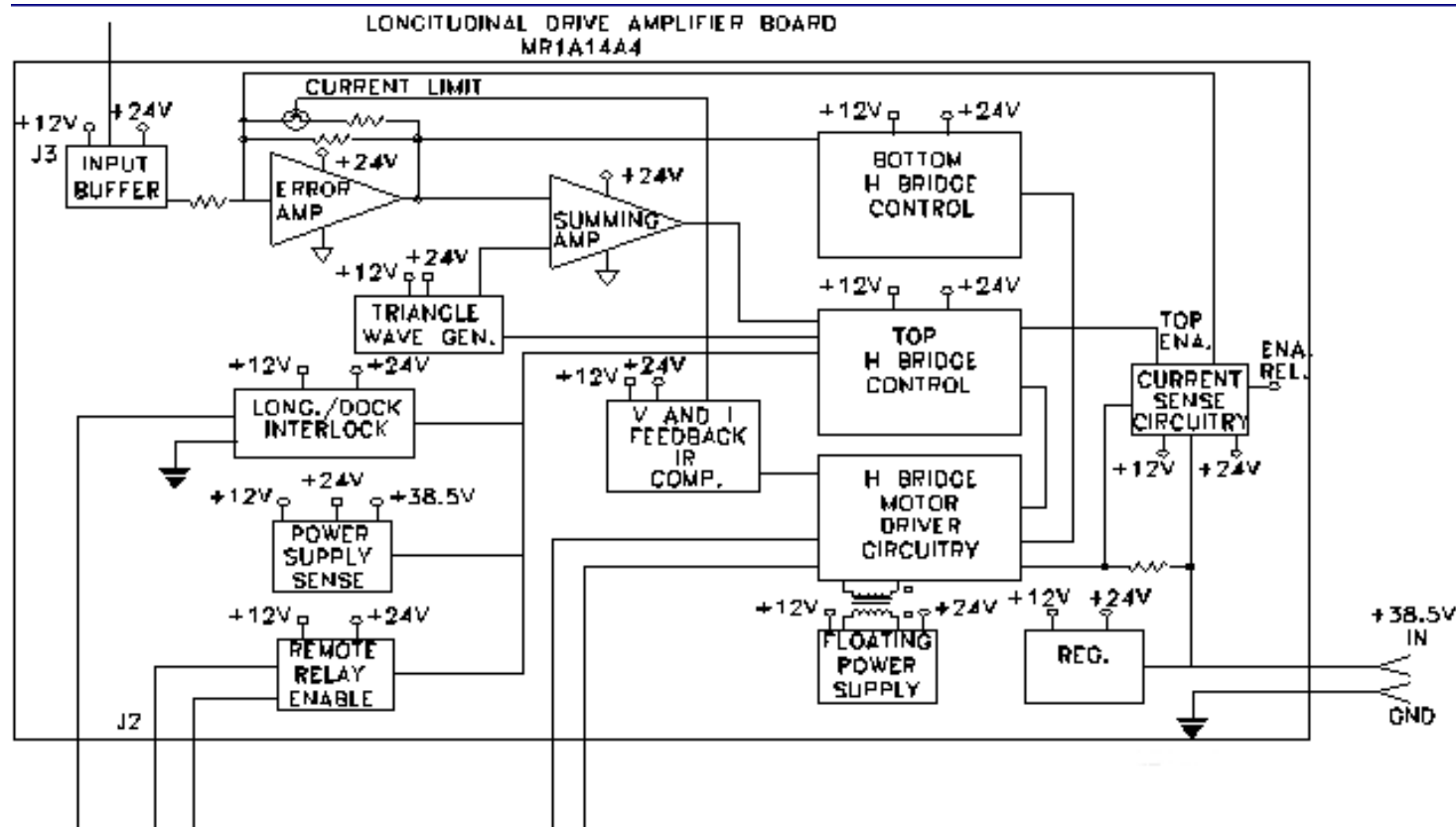
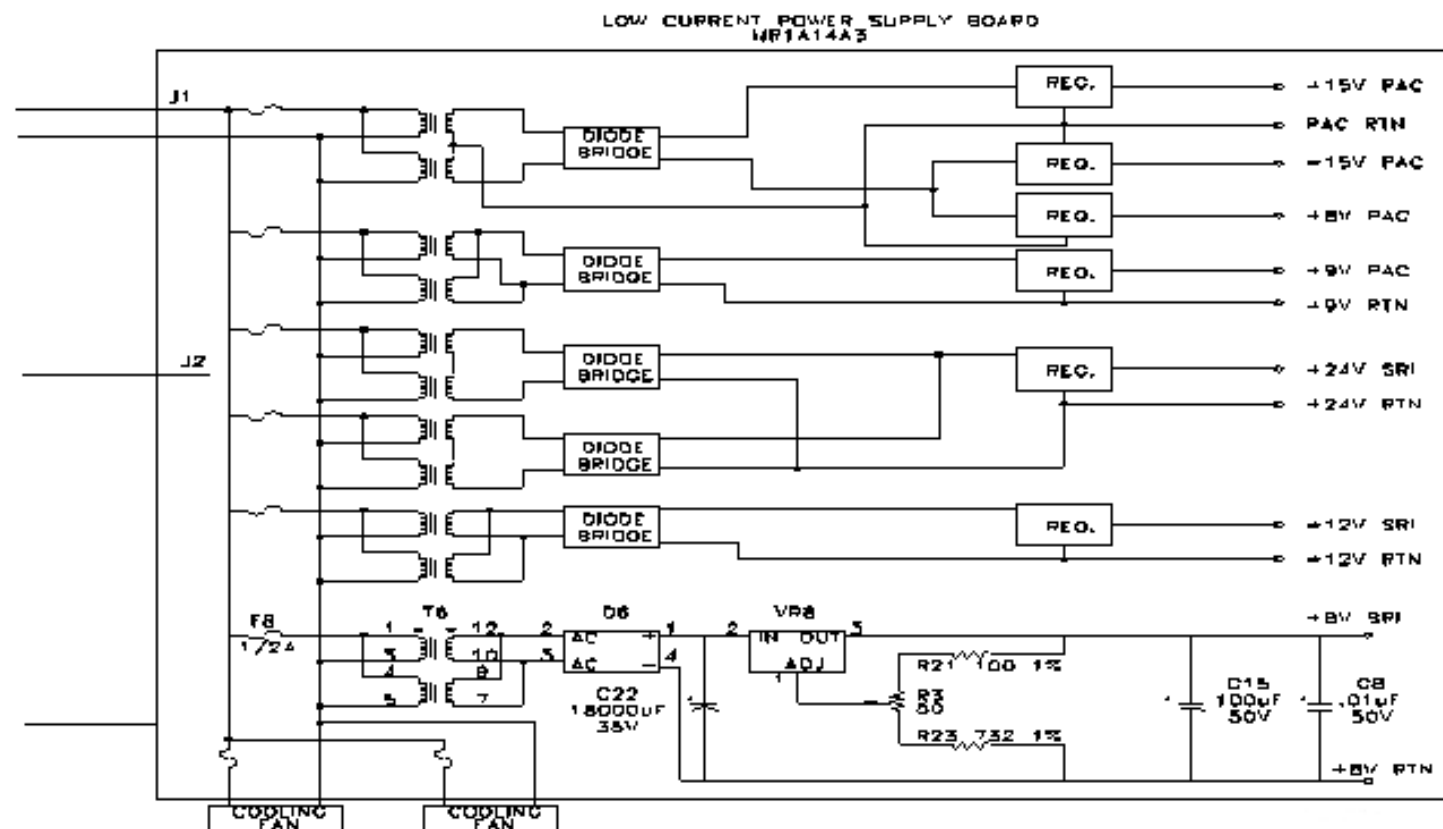


ILLUSTRATION L3185A
MEPS FUNCTIONAL BLOCK DIAGRAM

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 27, 1998	M. Whitlow	Initial conversion from Toolbook to Word.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document, fixed table of contents.